EE3005 Image Processing (IP) Assignment 1

IP Assignment 1: Fundamentals of Image Representation and Enhancement

Objective:

The primary goal of this assignment is to introduce students to fundamental concepts and practical techniques in image processing. Students will gain hands-on experience with image representation, RGB channel separation, DICOM image reading and metadata extraction, point operations, histogram analysis, and histogram equalization. Through these tasks, students will learn how pixel values and their distributions can be analyzed and manipulated to improve image quality and visual information. The assignment also aims to develop students ability to implement image processing techniques in MATLAB, interpret their results, and critically compare different image enhancement approaches.

Task 1: Image Reading and Matrix Representation

1. Image Reading:

- Choose a grey level biomedical image of your choice (ensure it is available in a common format like JPEG or PNG).

- Use MATLAB programming language to read the image into your code.

2. Matrix Representation:

- Once the image is loaded, represent it as a matrix. Explain how each pixel in the image corresponds to an element in the matrix and how the color information is encoded.

3. Display Matrix:

- Write a function to display the matrix representation of the image. This should provide a clear visual of how the image is represented as a collection of numerical values.

Task 2: Separating RGB Channels

1. RGB Image:

- Explain the concept of RGB images, where each pixel is a combination of Red, Green, and Blue components.

2. Separation of Channels:

- Choose a biomedical image and implement a function to separate the RGB image into its individual Red, Green, and Blue channels. Discuss how this separation is achieved using matrix operations.

3. Display Individual Channels:

- Display each separated channel as a grayscale image. Discuss the visual characteristics of each channel and how they contribute to the overall color of the original image.

Task 3: Summary and Analysis

1. Write-up:

- In a brief summary, explain the importance of understanding image representation as matrices and the significance of separating RGB channels in image processing.

2. Applications:

- Discuss potential real-world applications of these image processing techniques. How might understanding image matrices and RGB channel separation be useful in fields such as computer vision, medical imaging, or multimedia processing?

Task 4: DICOM Image Reading, Display, and Metadata Extraction

1. DICOM Image Reading:

- Obtain a DICOM (Digital Imaging and Communications in Medicine) format medical image. Explain the significance of DICOM in medical imaging and how it differs from standard image formats.

2. Read and Display DICOM Image:

- Use MATLAB programming language with DICOM support to read and display the DICOM image. Highlight any specific considerations or challenges associated with DICOM format.

3. Metadata Extraction:

- Develop a function to extract and display relevant metadata from the DICOM image. Include information such as patient details, acquisition parameters, and any other pertinent data.

Task 5: Point Operations

1. Brightness Adjustment:

Select a grayscale biomedical image and apply a point operation to increase and decrease its brightness.

Explain how adding a constant value to pixel intensities affects the image.

2. Contrast Adjustment:

Apply different gain values to the image to increase and decrease its contrast.

Explain the relationship between the gain parameter and image contrast.

3. Image Negative:

Generate the negative of the original grayscale image using a point transformation.

Compare the original and negative images and explain the observed changes.

4. Analysis:

Display the original and processed images together.

Discuss how brightness and contrast modifications affect the visual appearance and pixel intensity distribution.

Task 6: Histogram Analysis

1. Grayscale Histogram:

Select a grayscale biomedical image and generate its histogram.

Explain what the x-axis and y-axis represent.

2. Histogram Interpretation:

Analyze the histogram of the selected image and identify whether the intensity values are concentrated in dark, bright, or intermediate regions.

3. RGB Histogram:

Select a color biomedical image and generate the individual Red, Green, and Blue histograms.

Compare the three histograms and explain how they relate to the colors present in the image.

4. Analysis:

Discuss how the histogram can be used to understand the brightness and contrast characteristics of an image.

Task 7: Histogram Equalization

1. Original Image and Histogram:

Select a grayscale biomedical image with limited or uneven contrast.

Display the original image and its histogram.

2. Histogram Equalization:

Apply global histogram equalization to the selected image.

Display the equalized image and its histogram.

3. Comparison:

Compare the original and equalized images visually.

Compare their histograms and explain how the distribution of intensity values has changed.

4. Analysis:

Explain why histogram equalization can improve the visibility of details in an image.

Discuss whether histogram equalization always produces a visually better image.

Task 8: Image Enhancement Comparison

1. Enhancement Using Point Operations:

Apply brightness and contrast adjustments to the selected image.

2. Enhancement Using Histogram Equalization:

Apply histogram equalization to the same image.

3. Comparison:

Compare the results obtained using point operations and histogram equalization.

4. Discussion:

Explain which method provides better visual enhancement for the selected image and justify your answer based on the image histogram and visual appearance.

Submission Guidelines:

- Sample images have been provided in the "Image Samples" folder. You may also use different images that you find on your own.

- The number of group members should be maximum 3. One submission is enough per group.

- The entire report should be in a single Word document, including MATLAB code and images.

- Include the images and the source code for each steps with their explanations.

- Write a comprehensive report discussing your observations, insights, and any challenges faced during the implementation.

- Your reports will be checked by an AI detector. You will receive a grade of zero if AI is detected. Even if you use it only for grammar correction, the AI detector may detect AI. For your information.

Note: This assignment aims to provide students with a well-rounded understanding of image processing, including common image formats, medical imaging considerations, and advanced techniques for contrast enhancement. Students will gain practical experience in implementing and analyzing various image processing tasks.

Grading Rubric – 100 Points

Criteria	Description	Points
1. Image Reading & Matrix Representation	Correct image reading, matrix representation, pixel interpretation, and appropriate MATLAB implementation.	**15**
2. RGB Channel Separation	Correct separation of RGB channels, visualization of individual channels, and interpretation of their characteristics.	**10**
3. Summary & Applications	Clear explanation of the importance of image representation and RGB separation, with relevant real-world applications.	**5**
4. DICOM & Metadata	Correct DICOM image reading and display, understanding of DICOM, and extraction and interpretation of relevant metadata.	**15**
5. Point Operations	Correct implementation of brightness adjustment, contrast adjustment, and image negative, with appropriate analysis.	**15**
6. Histogram Analysis	Correct generation and interpretation of grayscale and RGB histograms, including brightness and contrast analysis.	**10**
7. Histogram Equalization	Correct application of global histogram equalization and meaningful comparison of original and equalized images and histograms.	**15**
8. Enhancement Comparison	Meaningful comparison between point operations and histogram equalization, supported by visual and histogram-based evidence.	**10**
9. Report Quality & Presentation	Organization, clarity, figures, MATLAB code, explanations, observations, and overall technical presentation.	**5**
Total		**100**